

CSE 102: Data Structures and Algorithms

MidSem Solution and Rubric (Section-A)

24 February 2026

Question 1

(a) Recursive Insertion Sort (10 Marks)

```
RECURSIVE-INSERTION-SORT( $A, n$ )
1  if  $n \leq 1$ 
2      return
3
4  RECURSIVE-INSERTION-SORT( $A, n-1$ )
5   $key = A[n]$ 
6   $i = n - 1$ 
7  while  $i > 0$  and  $A[i] > key$ 
8       $A[i+1] = A[i]$ 
9       $i = i - 1$ 
10
11  $A[i+1] = key$ 
```

Rubric (10 Marks)

- Correct recursive logic and insertion mechanism (5)
- Correct base case and termination (2)
- Correct indexing/boundary handling (2)
- Clarity, consistency, and structure (1)

Note: Any logically correct recursive formulation of insertion sort will be accepted even if syntactically different from the model solution.

(b) Recurrence and Recursion Tree (5 Marks)

Recurrence:

$$T(n) = T(n - 1) + cn$$

Recursion Tree:

$$\begin{array}{c} T(n) \quad \rightarrow \text{work: } cn \\ | \\ T(n-1) \quad \rightarrow \text{work: } c(n-1) \\ | \\ T(n-2) \quad \rightarrow \text{work: } c(n-2) \\ | \\ \vdots \\ | \\ T(1) \quad \rightarrow \text{Base case: } c \end{array}$$

At each level, the non-recursive work is:

$$cn, c(n-1), c(n-2), \dots, c$$

Total work:

$$\begin{aligned} T(n) &= c(n + (n-1) + (n-2) + \dots + 1) \\ &= c \frac{n(n+1)}{2} \\ T(n) &= \Theta(n^2) \end{aligned}$$

Rubric (5 Marks)

- Correct recursion tree (2)
- Proper recurrence expansion (2)
- Correct final complexity (1)

Question 2 (10 Marks)

Inductive hypothesis: Assume $T(k) \leq ck^2$ for $k < n$:

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + n \\ &\leq 4 \cdot c \cdot \frac{n^2}{4} + n \\ &= cn^2 + n \\ &= cn^2 - (-n) \\ T(n) &\leq cn^2 \text{ for no choice of } c > 0. \end{aligned}$$

Strengthening the inductive hypothesis. We subtract the lower-order term.

Inductive hypothesis: Assume that $T(k) \leq cn^2 - dn$ for $k < n$:

$$\begin{aligned} T(n) &= 4 \cdot T\left(\frac{n}{2}\right) + n \\ &\leq 4 \left(c \cdot \frac{n^2}{4} - \frac{dn}{2} \right) + n \\ &= cn^2 - 2dn + n \\ &= cn^2 - dn - (dn - n) \quad (\text{desired} - \text{residual}) \\ T(n) &\leq cn^2 - dn, \quad \text{whenever } dn - n \geq 0 \Rightarrow d - 1 \geq 0 \Rightarrow d \geq 1 \text{ and } n \geq 1 \end{aligned}$$

Hence,

$$T(n) = O(n^2) \quad \text{---(1)}$$

To prove $T(n) = \Theta(n^2)$, we also need to prove the lower bound.

Inductive Hypothesis: Assume that $T(k) \geq ck^2$ for $k < n$:

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + n \\ &\geq 4 \cdot c \cdot \frac{n^2}{4} + n \\ &= cn^2 + n \\ T(n) &\geq cn^2, \quad \forall c > 0 \text{ and } n \geq 1 \end{aligned}$$

Hence,

$$T(n) = \Omega(n^2) \quad \text{---(2)}$$

$$(1) \ \& \ (2) \Rightarrow \quad T(n) = \Theta(n^2).$$

Rubric (10 Marks)

- Subtracting lower-order term to strengthen hypothesis (2)
- Correct upper-bound hypothesis (3)
- Correct lower-bound hypothesis (3)
- Proper substitution and simplification (2)

Question 3 (10 Marks)

Asymptotic Time Complexity

Asymptotic time complexity describes the growth rate of an algorithm's running time as input size n approaches infinity.

Big-O Notation

Asymptotic Definition

$$f(n) = O(g(n))$$

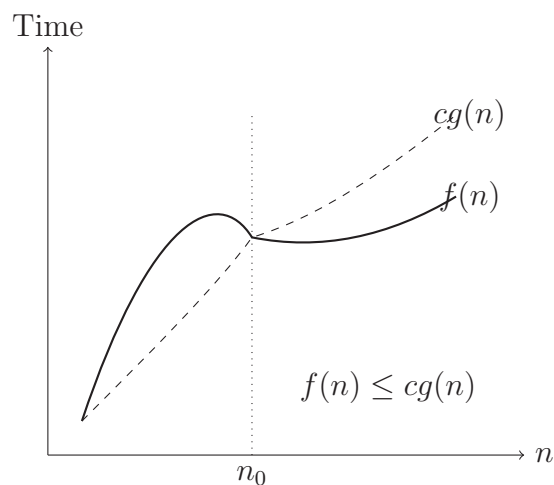
if there exist constants $c > 0$ and $n_0 > 0$ such that:

$$0 \leq f(n) \leq cg(n), \quad \forall n \geq n_0$$

Set Definition

$$O(g(n)) = \{f(n) \mid \exists c > 0, \exists n_0 > 0 \text{ such that } 0 \leq f(n) \leq cg(n), \forall n \geq n_0\}$$

Graphical Interpretation of $O(g(n))$



Big-Omega Notation

Asymptotic Definition

$$f(n) = \Omega(g(n))$$

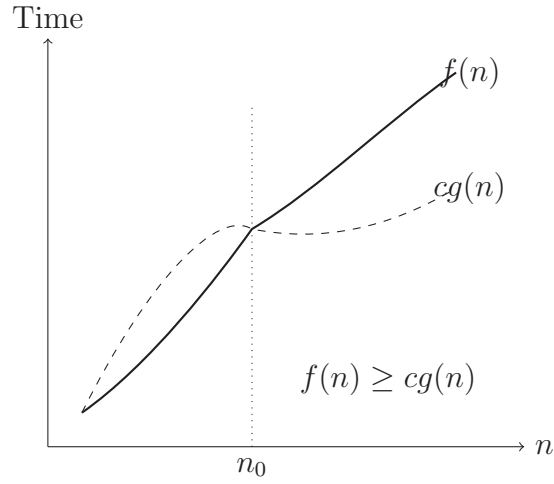
if there exist constants $c > 0$ and $n_0 > 0$ such that:

$$0 \leq cg(n) \leq f(n), \quad \forall n \geq n_0$$

Set Definition

$$\Omega(g(n)) = \{f(n) \mid \exists c > 0, \exists n_0 > 0 \text{ such that } 0 \leq cg(n) \leq f(n), \forall n \geq n_0\}$$

Graphical Interpretation of $\Omega(g(n))$



Big-Theta Notation

Asymptotic Definition

$$f(n) = \Theta(g(n))$$

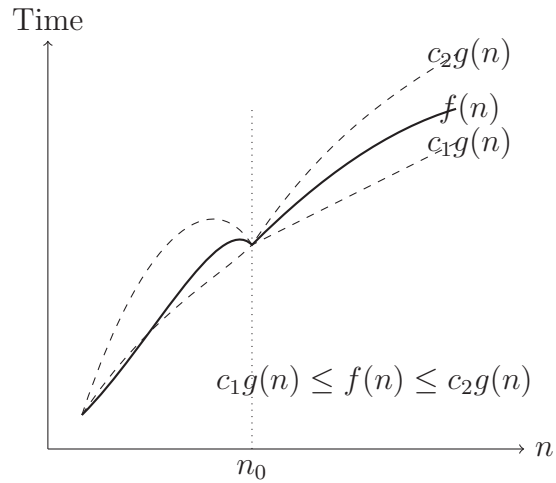
if there exist constants $c_1, c_2 > 0$ and n_0 such that:

$$0 \leq c_1g(n) \leq f(n) \leq c_2g(n), \quad \forall n \geq n_0$$

Set Definition

$$\Theta(g(n)) = \{f(n) \mid \exists c_1, c_2 > 0, \exists n_0 \text{ such that } 0 \leq c_1g(n) \leq f(n) \leq c_2g(n), \forall n \geq n_0\}$$

Graphical Interpretation of $\Theta(g(n))$



Rubric (10 Marks)

- Definition of asymptotic time complexity (1)
- Correct asymptotic or set definition of O (2.5)
- Correct asymptotic or set definition of Ω (2.5)
- Correct asymptotic or set definition of Θ (2.5)
- Correct graphical interpretations (1.5)

Note: Either the asymptotic definition or the equivalent set-based definition is acceptable for full credit.

Question 4

(a) Counting Sort Pseudocode (10 Marks)

Note: Multiple logically equivalent implementations of Counting Sort are acceptable. The three formulations provided below (zero-based indexing, one-based indexing, and range-based to handle negative keys) are all correct, provided the indexing is internally consistent and the algorithm correctly performs counting, prefix sums, and stable placement.

COUNTING-SORT-ZERO-BASED(A, n, k)

```
1  let  $B[1 \dots n]$  and  $C[0 \dots k]$  be new arrays
2  for  $i = 0$  to  $k$ 
3       $C[i] = 0$ 
4
5  for  $j = 1$  to  $n$ 
6       $C[A[j]] = C[A[j]] + 1$ 
7
8  for  $i = 1$  to  $k$ 
9       $C[i] = C[i] + C[i - 1]$ 
10
11 for  $j = n$  downto 1
12      $B[C[A[j]]] = A[j]$ 
13      $C[A[j]] = C[A[j]] - 1$ 
14
15 return  $B$ 
```

COUNTING-SORT-ONE-BASED(A, n, k)

```
1  let  $B[1 \dots n]$  and  $C[1 \dots k]$  be new arrays
2  for  $i = 1$  to  $k$ 
3       $C[i] = 0$ 
4
5  for  $j = 1$  to  $n$ 
6       $C[A[j]] = C[A[j]] + 1$ 
7
8  for  $i = 2$  to  $k$ 
9       $C[i] = C[i] + C[i - 1]$ 
10
11 for  $j = n$  downto 1
12      $B[C[A[j]]] = A[j]$ 
13      $C[A[j]] = C[A[j]] - 1$ 
14
15 return  $B$ 
```

COUNTING-SORT-RANGE-BASED(A, n)

```

1   $m = \min(A[1 \dots n])$ 
2   $M = \max(A[1 \dots n])$ 
3  let  $B[1 \dots n]$  and  $C[0 \dots (M - m)]$  be new arrays
4  for  $i = 0$  to  $M - m$ 
5       $C[i] = 0$ 
6
7  for  $j = 1$  to  $n$ 
8       $C[A[j] - m] = C[A[j] - m] + 1$ 
9
10 for  $i = 1$  to  $M - m$ 
11      $C[i] = C[i] + C[i - 1]$ 
12
13 for  $j = n$  downto 1
14      $B[C[A[j] - m]] = A[j]$ 
15      $C[A[j] - m] = C[A[j] - m] - 1$ 
16
17 return  $B$ 

```

Rubric (10 Marks)

- Correct array initialization (1)
- Frequency counting (4)
- Prefix sum computation (4)
- Consistency and structure (1)

Note: The Count array (C) can be initialized as either $C[0 \dots k]$ or $C[1 \dots k]$, to store counts for each integer value from $0 \dots k$ or $1 \dots k$ respectively.

Any logical formulation of counting sort may be accepted even if different from the model solutions.

Marks should not be deducted for choice of indexing convention, provided:

- The indexing is internally consistent,
- The key range assumption is clearly stated or logically implied,
- The algorithm correctly performs counting, prefix sum, and stable placement during backward traversal.

(b) Illustration (5 Marks)

We follow the CLRS 1-based indexing convention.

$$A[1 \dots 11] = \langle 6, 0, 2, 0, 1, 3, 4, 6, 1, 3, 2 \rangle$$

Here:

$$n = 11, \quad k = 6$$

We use:

$$C[0 \dots 6], \quad B[1 \dots 11]$$

Step 1: Frequency Counting

$$C = \langle 2, 2, 2, 2, 1, 0, 2 \rangle$$

Step 2: Cumulative Counts

$$C[0] = 2$$

$$C[1] = 2 + 2 = 4$$

$$C[2] = 4 + 2 = 6$$

$$C[3] = 6 + 2 = 8$$

$$C[4] = 8 + 1 = 9$$

$$C[5] = 9 + 0 = 9$$

$$C[6] = 9 + 2 = 11$$

Thus,

$$C = \langle 2, 4, 6, 8, 9, 9, 11 \rangle$$

Step 3: Backward Traversal

We traverse from $j = 11$ down to 1.

$$A[11] = 2 \Rightarrow B[6] = 2, \quad C[2] = 6 \rightarrow 5$$

$$A[10] = 3 \Rightarrow B[8] = 3, \quad C[3] = 8 \rightarrow 7$$

$$A[9] = 1 \Rightarrow B[4] = 1, \quad C[1] = 4 \rightarrow 3$$

$$A[8] = 6 \Rightarrow B[11] = 6, \quad C[6] = 11 \rightarrow 10$$

$$A[7] = 4 \Rightarrow B[9] = 4, \quad C[4] = 9 \rightarrow 8$$

$$A[6] = 3 \Rightarrow B[7] = 3, \quad C[3] = 7 \rightarrow 6$$

$$A[5] = 1 \Rightarrow B[3] = 1, \quad C[1] = 3 \rightarrow 2$$

$$A[4] = 0 \Rightarrow B[2] = 0, \quad C[0] = 2 \rightarrow 1$$

$$A[3] = 2 \Rightarrow B[5] = 2, \quad C[2] = 5 \rightarrow 4$$

$$A[2] = 0 \Rightarrow B[1] = 0, \quad C[0] = 1 \rightarrow 0$$

$$A[1] = 6 \Rightarrow B[10] = 6, \quad C[6] = 10 \rightarrow 9$$

Final Sorted Output

$$B[1 \dots 11] = \langle 0, 0, 1, 1, 2, 2, 3, 3, 4, 6, 6 \rangle$$

Rubric (5 Marks)

- Correct array ranges (1)
- Correct frequency counts (1)
- Correct cumulative counts (1)
- Detailed intermediate steps (2)